

Exercise J02

Q1. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as $\frac{m}{n}$ then find the value of $(m + n)$.

Ans. 191

Sol. -

Q2. A quadratic polynomial $f(x) = x^2 + ax + b$ is formed with one of its zeros being $\frac{4 + 3\sqrt{3}}{2 + \sqrt{3}}$ where a and b are integers. Also $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ is a biquadratic polynomial such that $g\left(\frac{4 + 3\sqrt{3}}{2 + \sqrt{3}}\right) = c\sqrt{3} + d$ where c and d are also integers. Find the values of a, b, c and d .

Ans. $a=2, b=-11, c=4, d=-1$

Sol. Rationalize the given zero:

$$\frac{4 + 3\sqrt{3}}{2 + \sqrt{3}} = \frac{(4 + 3\sqrt{3})(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} = \frac{8 - 4\sqrt{3} + 6\sqrt{3} - 9}{1} = -1 + 2\sqrt{3}.$$

Since the coefficients are integers, the other zero must be the conjugate $-1 - 2\sqrt{3}$.

Using Vieta's formulas, sum of zeros $= -a = (-1 + 2\sqrt{3}) + (-1 - 2\sqrt{3}) = -2$, so $a = 2$.

Product $= b = (-1 + 2\sqrt{3})(-1 - 2\sqrt{3}) = 1 - 12 = -11$. $f(x) = x^2 + 2x - 11$,

For $g(x)$, perform polynomial long division of $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ by $f(x) = x^2 + 2x - 11$.

The division gives quotient $x^2 + 1$ and remainder $2x + 1$.

Thus $g(x) = (x^2 + 1)f(x) + (2x + 1)$. Since $f(x_0) = 0$, $g(x_0) = 2x_0 + 1$.

Substitute the simplified zero $x_0 = -1 + 2\sqrt{3}$ (obtained by rationalizing) to get $2(-1 + 2\sqrt{3}) + 1 = 4\sqrt{3} - 1$, so $c = 4, d = -1$.

Q3. α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

Ans. 73

Sol. $\therefore \alpha, \beta$ are roots of $\lambda x^2 - (\lambda - 1)x + 5 = 0 \quad \therefore \alpha + \beta = \frac{\lambda - 1}{\lambda}$ and $\alpha\beta = \frac{5}{\lambda}$

$$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{30}{\lambda} \Rightarrow \frac{\alpha^2 + \beta^2}{\alpha\beta} = 4 \Rightarrow (\alpha + \beta)^2 = 6\alpha\beta$$

$$\Rightarrow \frac{(\lambda - 1)^2}{\lambda^2} = \frac{30}{\lambda} \Rightarrow \lambda^2 - 32\lambda + 1 = 0 \dots\dots (1)$$

$$\therefore \frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} = \frac{(\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2}{\lambda_1\lambda_2} = \frac{(32)^2 - 2}{1} = 1022 \Rightarrow \left(\frac{\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}}{14} \right) = 73$$

Q4. If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b + c + 1 = 0$ or $b^2 + c^2 + 1 = bc + b + c$.

Ans. -

Sol. Let α be the common root of the two quadratic equations.

Then α satisfies both equations:

$$\alpha^2 + b\alpha + c = 0 \text{ ——— (1)}$$

$$b\alpha^2 + c\alpha + 1 = 0 \text{ ——— (2)}$$

Multiply equation (1) by b to obtain:

$$b\alpha^2 + b^2\alpha + bc = 0 \text{ ——— (3)}$$

Subtract equation (3) from equation (2):

$$(b\alpha^2 + c\alpha + 1) - (b\alpha^2 + b^2\alpha + bc) = 0$$

This simplifies to:

$$(c - b^2)\alpha + (1 - bc) = 0$$

Hence, when $c \neq b^2$, we can solve for α : $\alpha = \frac{bc - 1}{c - b^2}$

Substitute this expression for α into equation (1): $\left(\frac{bc - 1}{c - b^2} \right)^2 + b \left(\frac{bc - 1}{c - b^2} \right) + c = 0$

Multiply through by $(c - b^2)^2$: $(bc - 1)^2 + b(bc - 1)(c - b^2) + c(c - b^2)^2 = 0$

Expand each term: $(bc - 1)^2 = b^2c^2 - 2bc + 1$

$$b(bc - 1)(c - b^2) = b^2c^2 - bc - b^4c + b^3$$

$$c(c - b^2)^2 = c^3 - 2b^2c^2 + b^4c$$

Summing these gives:

$$(b^2c^2 - 2bc + 1) + (b^2c^2 - bc - b^4c + b^3) + (c^3 - 2b^2c^2 + b^4c) = 0$$

Cancel like terms: the b^2c^2 terms cancel, the $-b^4c$ and $+b^4c$ cancel, leaving: $c^3 + b^3 - 3bc + 1 = 0$

This expression factorises as: $(b + c + 1)(b^2 + c^2 + 1 - bc - b - c) = 0$

Thus either $b + c + 1 = 0$ or $b^2 + c^2 + 1 - bc - b - c = 0$, which is equivalent to $b^2 + c^2 + 1 = bc + b + c$.

Q5. Find the value of m for which the quadratic equations $x^2 - 11x + m = 0$ & $x^2 - 14x + 2m = 0$ may have common root.

Ans. 0, 24

Sol. Given equation are $x^2 - 11x + a = 0$ (i) $x^2 - 14x + 2a = 0$ (ii)
 Multiplying equation (i) by 2 and then subtracting, we get $x^2 - 8x = 0 \Rightarrow x = 0, 8$
 If $x = 0$, $a = 0$ If $x = 8$, $a = 24$

Q6. Find the product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and
 $Q(x) = x^3 + 4x^2 - x - 10$.

Ans. 6

Sol. Considering denominator $x^2 - 8x + 32$

$$D < 0 \text{ and } a > 0$$

So denominator is always positive

$$\Rightarrow ax^2 + 2(a+1)x + 9a + 4 < 0$$

$$\Rightarrow a < 0 \text{ \& } 4(a+1)^2 - 4a(9a+4) < 0$$

$$\Rightarrow 4(a^2 + 2a + 1 - 9a^2 - 4a) < 0$$

$$\Rightarrow 4(-8a^2 - 2a + 1) < 0$$

$$8a^2 + 2a - 1 > 0$$

$$(4a - 1)(2a + 1) > 0$$

$$\Rightarrow a \in \left(-\infty, -\frac{1}{2}\right)$$

Q7. Let α, β be real roots of the quadratic equation $x^2 - kx + k^2 + k - 5 = 0$. If m and M are respectively the minimum and maximum value of $\alpha^2 + \beta^2$, then find $(m + M)$.

Ans. 13

Sol. Let the quadratic equation be $x^2 - kx + (k^2 + k - 5) = 0$.

The sum of the roots α and β is $\alpha + \beta = k$.

The product of the roots is $\alpha\beta = k^2 + k - 5$.

We need $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = k^2 - 2(k^2 + k - 5)$.

Simplify to get $\alpha^2 + \beta^2 = -k^2 - 2k + 10 = 11 - (k + 1)^2$.

For real roots, the discriminant must be non-negative: $(-k)^2 - 4 \cdot 1 \cdot (k^2 + k - 5) \geq 0$.

This gives $k^2 - 4k^2 - 4k + 20 \geq 0$, so $-3k^2 - 4k + 20 \geq 0$.

Multiply by -1 to get $3k^2 + 4k - 20 \leq 0$.

The roots of $3k^2 + 4k - 20 = 0$ are $k = \frac{-4 \pm \sqrt{16 + 240}}{6} = \frac{-4 \pm 16}{6}$, giving $k = 2$ and $k = -\frac{10}{3}$.

Thus the inequality holds for $k \in \left[-\frac{10}{3}, 2\right]$.

Now $\alpha^2 + \beta^2 = 11 - (k + 1)^2$. This is a downward parabola in k , so its maximum on the interval occurs at the vertex $k = -1$, which lies inside $\left[-\frac{10}{3}, 2\right]$.

At $k = -1$, the value is $11 - 0 = 11$. Hence the maximum $M = 11$.

The minimum on the closed interval will occur at an endpoint.

$$\text{At } k = -\frac{10}{3}, \alpha^2 + \beta^2 = 11 - \left(-\frac{10}{3} + 1\right)^2 = 11 - \left(-\frac{7}{3}\right)^2 = 11 - \frac{49}{9} = \frac{50}{9}.$$

$$\text{At } k = 2, \alpha^2 + \beta^2 = 11 - (2 + 1)^2 = 11 - 9 = 2.$$

The smaller value is 2, so the minimum $m = 2$.

Therefore, $m + M = 2 + 11 = 13$.

Q8. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ has unequal real roots for all $a \in \mathbb{R}$.

Ans. 5

Sol. We are given the quadratic equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ with arbitrary real numbers a and b .

For the quadratic to have unequal real roots, the discriminant must be strictly positive.

The discriminant D is given by $D = [2(a + b)]^2 - 4 \cdot 1 \cdot (a - b + 8)$.

$$\text{Simplify: } D = 4(a + b)^2 - 4(a - b + 8) = 4[(a + b)^2 - (a - b + 8)].$$

Thus the condition $D > 0$ is equivalent to $(a + b)^2 - (a - b + 8) > 0$.

$$\text{Expand: } (a + b)^2 - a + b - 8 = a^2 + 2ab + b^2 - a + b - 8.$$

Treat this as a quadratic in a : $a^2 + (2b - 1)a + (b^2 + b - 8) > 0$ for all real a .

Since the coefficient of a^2 is $1 > 0$, this quadratic is positive for all a if and only if its discriminant is negative.

$$\text{Let } D_a \text{ be the discriminant of the quadratic in } a: D_a = (2b - 1)^2 - 4 \cdot 1 \cdot (b^2 + b - 8).$$

$$\text{Compute: } (2b - 1)^2 = 4b^2 - 4b + 1 \text{ and } 4(b^2 + b - 8) = 4b^2 + 4b - 32.$$

$$\text{Thus } D_a = (4b^2 - 4b + 1) - (4b^2 + 4b - 32) = -8b + 33.$$

We require $D_a < 0$, i.e., $33 - 8b < 0$.

$$\text{This inequality gives } 8b > 33, \text{ or } b > \frac{33}{8} = 4.125.$$

Since b is a natural number (positive integer), the smallest such b is 5.

Therefore, the smallest natural number b for which the equation has unequal real roots for all real a is 5.

Q9. Find the range of values of a , such that $f(x) = \frac{ax^2 + 2(a + 1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

Ans. $a \in (-\infty, -\frac{1}{2})$

Sol. We need the range of a such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

First, note that the denominator $x^2 - 8x + 32 = (x - 4)^2 + 16 > 0$ for all real x , so the sign of $f(x)$ is determined solely by the numerator.

Thus, we require the numerator $N(x) = ax^2 + 2(a+1)x + 9a + 4 < 0$ for all real x .

For a quadratic to be negative for all x , the leading coefficient must be negative and the discriminant must be negative.

Hence, we need $a < 0$ and $D < 0$.

Compute the discriminant:
 $D = [2(a+1)]^2 - 4 \cdot a \cdot (9a+4) = 4(a+1)^2 - 4a(9a+4) = 4[(a+1)^2 - a(9a+4)]$.

Simplify: $(a+1)^2 - a(9a+4) = a^2 + 2a + 1 - 9a^2 - 4a = -8a^2 - 2a + 1$.

Thus, $D = 4(-8a^2 - 2a + 1)$.

Require $D < 0$, i.e., $-8a^2 - 2a + 1 < 0$.

Multiply both sides by -1 (reversing inequality): $8a^2 + 2a - 1 > 0$.

Solve the quadratic equation $8a^2 + 2a - 1 = 0$:

$$a = \frac{-2 \pm \sqrt{4 + 32}}{16} = \frac{-2 \pm 6}{16}, \text{ giving } a = \frac{1}{4} \text{ or } a = -\frac{1}{2}.$$

The inequality $8a^2 + 2a - 1 > 0$ holds for $a < -\frac{1}{2}$ or $a > \frac{1}{4}$.

Combining with $a < 0$, we get $a < -\frac{1}{2}$.

Now check the boundary $a = -\frac{1}{2}$: then numerator becomes $-\frac{1}{2}x^2 + x - \frac{1}{2} = -\frac{1}{2}(x-1)^2$, which is zero at $x = 1$, so not always negative.

Therefore, the required range is $a \in (-\infty, -\frac{1}{2})$.

Q10. We call 'p' a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ is satisfied for all real x .
 Find the smallest integral good number.

Ans. 4

Sol. We need the smallest integral good number p such that $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ for all real x .

$$\text{Let } y = \frac{2x^2 + 2x + 3}{x^2 + x + 1}.$$

Since the denominator $x^2 + x + 1$ is always positive (discriminant $1 - 4 = -3 < 0$ and leading coefficient positive), we can multiply both sides by it.

This gives $y(x^2 + x + 1) = 2x^2 + 2x + 3$.

Rearrange: $yx^2 + yx + y = 2x^2 + 2x + 3$.

Bring all terms to one side: $(y - 2)x^2 + (y - 2)x + (y - 3) = 0$.

For x to be real, the discriminant of this quadratic must be nonnegative:

$$\Delta = (y - 2)^2 - 4(y - 2)(y - 3) \geq 0.$$

$$\text{Factor: } (y - 2)[(y - 2) - 4(y - 3)] = (y - 2)(y - 2 - 4y + 12) = (y - 2)(-3y + 10) \geq 0.$$

Multiply by -1 (reversing inequality): $(y - 2)(3y - 10) \leq 0$.

$$\text{Thus } y \in [2, \frac{10}{3}].$$

The maximum value of $\frac{2x^2 + 2x + 3}{x^2 + x + 1}$ is $\frac{10}{3} \approx 3.33$.

The smallest integer p satisfying $p \geq \frac{10}{3}$ is 4.

Therefore, the smallest integral good number is $\boxed{4}$.

Q11. Let $x^2 + y^2 + xy + 1 \geq a(x + y) \forall x, y \in \mathbb{R}$. Find the possible integer(s) in the range of a .

Ans. -1, 0, 1

Sol. We need to find all integer values of a such that $x^2 + y^2 + xy + 1 \geq a(x + y)$ holds for all real x, y .

Rewrite the inequality as $x^2 + y^2 + xy + 1 - a(x + y) \geq 0$.

Treat this as a quadratic in x : $x^2 + (y - a)x + (y^2 - ay + 1) \geq 0$ for all x .

Since the coefficient of x^2 is positive ($1 > 0$), the quadratic is always nonnegative if its discriminant is ≤ 0 .

Compute the discriminant in x : $\Delta_x = (y - a)^2 - 4 \cdot 1 \cdot (y^2 - ay + 1)$.

$$\text{Simplify: } \Delta_x = (y^2 - 2ay + a^2) - 4y^2 + 4ay - 4 = -3y^2 + 2ay + (a^2 - 4).$$

We require $\Delta_x \leq 0$ for all real y . This is a quadratic in y : $-3y^2 + 2ay + (a^2 - 4) \leq 0$ for all y .

The leading coefficient is negative ($-3 < 0$), so this quadratic is always nonpositive if its discriminant (in y) is ≤ 0 .

Compute the discriminant in y :
 $\Delta_y = (2a)^2 - 4(-3)(a^2 - 4) = 4a^2 + 12(a^2 - 4) = 16a^2 - 48 = 16(a^2 - 3)$.

Set $\Delta_y \leq 0$: $16(a^2 - 3) \leq 0$ implies $a^2 - 3 \leq 0$, so $a^2 \leq 3$.

$$\text{Thus } -\sqrt{3} \leq a \leq \sqrt{3}.$$

The integer values in this interval are $-1, 0, 1$.

Q12. Show that the equation $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$ for every real value of x , there is a real value of y and for every real value of y there is a real value of x .

Ans. -

Sol. We are given the equation $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$.

We need to show that for every real value of x , there is a real value of y , and for every real value of y , there is a real value of x .

First, treat the equation as a quadratic in y . Rearrange: $2y^2 - (3x + 3)y + (x^2 - 2x - 35) = 0$.

For y to be real, the discriminant of this quadratic must be non-negative.

Compute the discriminant Δ_y : $\Delta_y = [-(3x + 3)]^2 - 4 \cdot 2 \cdot (x^2 - 2x - 35)$.

$$= x^2 + 34x + 289. \quad x^2 + 34x + 289 = (x + 17)^2.$$

Since $(x + 17)^2 \geq 0$ for all real x , we have $\Delta_y \geq 0$ for every real x .

Hence, for each real x , the quadratic in y has real roots, so there exists a real y satisfying the equation.

Now, treat the equation as a quadratic in x . Rearrange: $x^2 - (3y + 2)x + (2y^2 - 3y - 35) = 0$.

For x to be real, the discriminant of this quadratic must be non-negative.

Compute the discriminant Δ_x : $\Delta_x = [-(3y + 2)]^2 - 4 \cdot 1 \cdot (2y^2 - 3y - 35)$.

$$= y^2 + 24y + 144.$$

Notice that $y^2 + 24y + 144 = (y + 12)^2$.

Since $(y + 12)^2 \geq 0$ for all real y , we have $\Delta_x \geq 0$ for every real y .

Hence, for each real y , the quadratic in x has real roots, so there exists a real x satisfying the equation.

Therefore, for every real x there is a real y , and for every real y there is a real x , satisfying the given equation.

Q13. If α, β and γ are the roots of the equation $5x^3 - qx - 1 = 0$, ($q \in \mathbb{R}$) then find the value of $\frac{\alpha^2 - 3}{\beta\gamma} + \frac{\beta^2 - 3}{\gamma\alpha} + \frac{\gamma^2 - 3}{\alpha\beta}$.

Ans. 3

Sol. Given the cubic equation $5x^3 - qx - 1 = 0$ with roots α, β, γ .

Comparing with $5x^3 + 0x^2 - qx - 1 = 0$, we have:

$$\alpha + \beta + \gamma = -\frac{0}{5} = 0,$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{-q}{5} = -\frac{q}{5},$$

$$\alpha\beta\gamma = -\frac{-1}{5} = \frac{1}{5}.$$

$$\text{We need to evaluate } S = \frac{\alpha^2 - 3}{\beta\gamma} + \frac{\beta^2 - 3}{\gamma\alpha} + \frac{\gamma^2 - 3}{\alpha\beta}.$$

Rewrite each term: $\frac{\alpha^2 - 3}{\beta\gamma} = \frac{\alpha^2}{\beta\gamma} - \frac{3}{\beta\gamma}$. Summing gives:

$$S = \sum \frac{\alpha^2}{\beta\gamma} - 3 \sum \frac{1}{\beta\gamma}.$$

Now, $\sum \frac{\alpha^2}{\beta\gamma} = \frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha\beta\gamma}$ because $\frac{\alpha^2}{\beta\gamma} = \frac{\alpha^3}{\alpha\beta\gamma}$.

Also, $\sum \frac{1}{\beta\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{0}{\alpha\beta\gamma} = 0$.

Thus, $S = \frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha\beta\gamma}$.

Since $\alpha + \beta + \gamma = 0$, we use the identity $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$.

Therefore, $\alpha^3 + \beta^3 + \gamma^3 = 3 \cdot \frac{1}{5} = \frac{3}{5}$.

Finally, $S = \frac{\frac{3}{5}}{\frac{1}{5}} = 3$.

Hence, the required value is 3.

Q14. Let α, β, γ are the roots of the cubic equation $x^3 + 2x^2 - x - 3 = 0$. If the absolute value of the expression $\frac{\alpha + 3}{\alpha - 2} + \frac{\beta + 3}{\beta - 2} + \frac{\gamma + 3}{\gamma - 2}$ can be expressed as $\frac{m}{n}$ where m and n are coprime, then find the value of $(m + n)$.

Ans. 73

Sol. Given the cubic equation $x^3 + 2x^2 - x - 3 = 0$ with roots α, β, γ .

We need to compute the expression $\frac{\alpha + 3}{\alpha - 2} + \frac{\beta + 3}{\beta - 2} + \frac{\gamma + 3}{\gamma - 2}$.

Rewrite each term: $\frac{\alpha + 3}{\alpha - 2} = \frac{(\alpha - 2) + 5}{\alpha - 2} = 1 + \frac{5}{\alpha - 2}$.

Thus the sum becomes $3 + 5 \left(\frac{1}{\alpha - 2} + \frac{1}{\beta - 2} + \frac{1}{\gamma - 2} \right)$.

Let $y = x - 2$. Then the roots in y are $\alpha - 2, \beta - 2, \gamma - 2$. Substitute $x = y + 2$ into the cubic:

$$(y + 2)^3 + 2(y + 2)^2 - (y + 2) - 3 = 0.$$

Expand: $(y^3 + 6y^2 + 12y + 8) + 2(y^2 + 4y + 4) - y - 2 - 3 = 0$.

Combine like terms: $y^3 + (6 + 2)y^2 + (12 + 8 - 1)y + (8 + 8 - 5) = y^3 + 8y^2 + 19y + 11 = 0$.

The new cubic is $y^3 + 8y^2 + 19y + 11 = 0$ with roots $y_1 = \alpha - 2, y_2 = \beta - 2, y_3 = \gamma - 2$.

For a cubic $y^3 + ay^2 + by + c = 0$, the sum of reciprocals of roots is $\sum \frac{1}{y_i} = -\frac{b}{c}$ provided $c \neq 0$. Here $a = 8, b = 19, c = 11$.

Therefore, $\frac{1}{\alpha - 2} + \frac{1}{\beta - 2} + \frac{1}{\gamma - 2} = -\frac{19}{11}$.

Substitute back into the expression: $S = 3 + 5 \left(-\frac{19}{11} \right) = 3 - \frac{95}{11} = \frac{33 - 95}{11} = -\frac{62}{11}$.

The absolute value is $\frac{62}{11}$. Thus $m = 62$ and $n = 11$ are coprime.

Finally, $m + n = 62 + 11 = 73$.

Q15. Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha - 2}, \frac{\beta}{\beta - 2}$ and $\frac{\gamma}{\gamma - 2}$. Hence or otherwise find the value of $(\alpha - 2)(\beta - 2)(\gamma - 2)$.

Ans. $3x^3 - 9x^2 - 3x + 1 = 0$; 3

Sol. We are given the cubic $x^3 - 3x^2 + 1 = 0$ with roots α, β, γ .

We need a cubic whose roots are $\frac{\alpha}{\alpha - 2}, \frac{\beta}{\beta - 2}, \frac{\gamma}{\gamma - 2}$.

Let $y = \frac{x}{x - 2}$.

Solve for x in terms of y : $y = \frac{x}{x - 2}$ implies $y(x - 2) = x$, so $yx - 2y = x$, then $yx - x = 2y$, giving $x(y - 1) = 2y$, hence $x = \frac{2y}{y - 1}$.

Substitute $x = \frac{2y}{y - 1}$ into the original cubic:

$$\left(\frac{2y}{y - 1}\right)^3 - 3\left(\frac{2y}{y - 1}\right)^2 + 1 = 0.$$

Multiply both sides by $(y - 1)^3$:

$$(2y)^3 - 3(2y)^2(y - 1) + (y - 1)^3 = 0.$$

$$\text{Compute: } 8y^3 - 3 \cdot 4y^2(y - 1) + (y - 1)^3 = 0.$$

$$\text{Expand: } 8y^3 - 12y^2(y - 1) + (y^3 - 3y^2 + 3y - 1) = 0.$$

$$\text{Simplify: } 8y^3 - 12y^3 + 12y^2 + y^3 - 3y^2 + 3y - 1 = 0.$$

$$\text{Combine like terms: } (8 - 12 + 1)y^3 + (12 - 3)y^2 + 3y - 1 = 0, \text{ so } -3y^3 + 9y^2 + 3y - 1 = 0.$$

$$\text{Multiply by } -1 \text{ to get the cubic with the same roots: } 3y^3 - 9y^2 - 3y + 1 = 0.$$

$$\text{Thus the required cubic is } 3x^3 - 9x^2 - 3x + 1 = 0.$$

$$\text{Now find } (\alpha - 2)(\beta - 2)(\gamma - 2).$$

$$\text{Let } t = x - 2. \text{ Then } x = t + 2.$$

$$\text{Substitute into the original equation: } (t + 2)^3 - 3(t + 2)^2 + 1 = 0.$$

$$\text{Expand: } t^3 + 6t^2 + 12t + 8 - 3(t^2 + 4t + 4) + 1 = 0.$$

$$\text{Simplify: } t^3 + 6t^2 + 12t + 8 - 3t^2 - 12t - 12 + 1 = 0.$$

$$\text{Combine: } t^3 + (6 - 3)t^2 + (12 - 12)t + (8 - 12 + 1) = 0, \text{ giving } t^3 + 3t^2 + 0t - 3 = 0.$$

$$\text{The roots of this cubic in } t \text{ are } \alpha - 2, \beta - 2, \gamma - 2.$$

$$\text{For a cubic } t^3 + at^2 + bt + c = 0, \text{ the product of roots is } -c.$$

$$\text{Here } c = -3, \text{ so product} = -(-3) = 3.$$

$$\text{Therefore, } (\alpha - 2)(\beta - 2)(\gamma - 2) = 3.$$

Q16. Two roots of a biquadratic $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ have their product equal to (-32) . Find the value of k .

Ans. $K=86$

Sol. Let the four roots be $\alpha, \beta, \gamma, \delta$.

By Vieta's formulas for the polynomial $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$, we have:

$$\alpha + \beta + \gamma + \delta = 18,$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = k,$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -200,$$

$$\alpha\beta\gamma\delta = -1984.$$

Given that the product of two roots is -32 . Without loss of generality, let $\alpha\beta = -32$.

From the product of all four roots: $(-32)(\gamma\delta) = -1984$ implies $\gamma\delta = 62$.

Let $S_1 = \alpha + \beta$ and $S_2 = \gamma + \delta$. Then from the sum of roots: $S_1 + S_2 = 18$.

From the sum of pairwise products: $\alpha\beta + (\alpha + \beta)(\gamma + \delta) + \gamma\delta = k$, i.e., $-32 + S_1S_2 + 62 = k$, so $k = S_1S_2 + 30$.

From the sum of triple products: $\alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = -200$, i.e., $(-32)S_2 + 62S_1 = -200$, giving $62S_1 - 32S_2 = -200$.

Solve the system:

$$S_1 + S_2 = 18 \text{ and } 62S_1 - 32S_2 = -200.$$

From the first, $S_2 = 18 - S_1$. Substitute into the second:

$$62S_1 - 32(18 - S_1) = -200 \text{ implies } 62S_1 - 576 + 32S_1 = -200 \text{ implies } 94S_1 = 376 \text{ implies } S_1 = 4.$$

$$\text{Then } S_2 = 18 - 4 = 14.$$

$$\text{Finally, } k = S_1S_2 + 30 = (4)(14) + 30 = 56 + 30 = 86.$$

Therefore, the value of k is 86.

Q17. Find the complete set of real values of 'a' for which both roots of the quadratic equation $(a^2 - 6a + 5)x^2 - \sqrt{a^2 + 2a}x + (6a - a^2 - 8) = 0$ lie on either side of the origin.

Ans. $a \in (-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$

Sol. We need to find all real a such that the quadratic equation $(a^2 - 6a + 5)x^2 - \sqrt{a^2 + 2a}x + (6a - a^2 - 8) = 0$ has both roots real and on opposite sides of the origin, i.e., one positive and one negative.

For a quadratic $Ax^2 + Bx + C = 0$, the product of roots is $\frac{C}{A}$. Roots have opposite signs if and only if $\frac{C}{A} < 0$, which is equivalent to $A \cdot C < 0$.

Here, $A = a^2 - 6a + 5 = (a - 1)(a - 5)$, $B = -\sqrt{a^2 + 2a}$, and $C = 6a - a^2 - 8 = -(a^2 - 6a + 8) = -(a - 2)(a - 4)$.

First, the square root requires $a^2 + 2a \geq 0 \Rightarrow a(a + 2) \geq 0 \Rightarrow a \leq -2$ or $a \geq 0$.

Second, the quadratic must be genuine: $A \neq 0$, so $a \neq 1$ and $a \neq 5$.

Third, the product condition: $A \cdot C < 0$. Compute $A \cdot C = (a^2 - 6a + 5)(6a - a^2 - 8) = -(a^2 - 6a + 5)(a^2 - 6a + 8)$. Let $t = a^2 - 6a$, then $A \cdot C = -(t + 5)(t + 8)$. The inequality $A \cdot C < 0$ becomes $-(t + 5)(t + 8) < 0 \Rightarrow (t + 5)(t + 8) > 0$. Hence $t < -8$ or $t > -5$.

Now $t = a^2 - 6a = (a - 3)^2 - 9 \geq -9$. Solve:

- $t < -8$ gives $a^2 - 6a + 8 < 0 \Rightarrow (a - 2)(a - 4) < 0 \Rightarrow 2 < a < 4$.
- $t > -5$ gives $a^2 - 6a + 5 > 0 \Rightarrow (a - 1)(a - 5) > 0 \Rightarrow a < 1$ or $a > 5$.

Thus the product condition gives $a \in (2, 4)$ or $a \in (-\infty, 1) \cup (5, \infty)$.

Now intersect with the domain condition $a \leq -2$ or $a \geq 0$:

- From $(2, 4)$: intersection with $a \geq 0$ gives $(2, 4)$.
- From $(-\infty, 1)$: intersection with $a \leq -2$ or $a \geq 0$ yields $(-\infty, -2] \cup [0, 1)$.
- From $(5, \infty)$: intersection with $a \geq 0$ gives $(5, \infty)$.

We must also exclude the points where $A = 0$ ($a = 1, 5$) which are already excluded (since $a = 1$ is not in $[0, 1)$ and $a = 5$ is not in $(5, \infty)$). We also check endpoints: at $a = -2$ and $a = 0$, the product is negative and roots are real (distinct), so they are included. At $a = 2$ and $a = 4$, one root is zero, which does not satisfy "opposite sides" (strictly positive/negative), so these points are excluded. Hence the final intervals are open at 2 and 4, closed at -2 and 0.

Therefore, the complete set of real a is:

$$a \in (-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty).$$

Q18. If α, β are the roots of the equation, $x^2 - 2x - a^2 + 1 = 0$ and γ, δ are the roots of the equation, $x^2 - 2(a + 1)x + a(a - 1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$ then find the values of 'a'.

Ans. $-\frac{1}{4} < a < 1$

Sol. Let the two quadratic equations be: $x^2 - 2x - a^2 + 1 = 0$ (roots α, β) and $x^2 - 2(a + 1)x + a(a - 1) = 0$ (roots γ, δ).

From the first equation, sum $\alpha + \beta = 2$ and product $\alpha\beta = 1 - a^2$. The roots are $1 \pm |a|$. From the second equation, sum $\gamma + \delta = 2(a + 1)$ and product $\gamma\delta = a(a - 1)$. The roots are $(a + 1) \pm \sqrt{3a + 1}$, which are real only when $a \geq -\frac{1}{3}$.

The condition $\alpha, \beta \in (\gamma, \delta)$ implies $\gamma < 1 - |a|$ and $1 + |a| < \delta$.

Consider two cases for a .

Case 1: $a \geq 0$. Then $|a| = a$. The inequalities become $a + 1 - \sqrt{3a + 1} < 1 - a$ and $1 + a < a + 1 + \sqrt{3a + 1}$. The first simplifies to $2a < \sqrt{3a + 1}$, which squaring gives $4a^2 < 3a + 1$ or $4a^2 - 3a - 1 < 0$. The roots of $4a^2 - 3a - 1 = 0$ are $a = 1$ and $a = -\frac{1}{4}$.

Hence $-\frac{1}{4} < a < 1$. With $a \geq 0$, we get $0 \leq a < 1$. The second inequality holds automatically for all $a \geq 0$.

Case 2: $-\frac{1}{3} \leq a < 0$ (for real γ, δ). Then $|a| = -a$. The first inequality becomes $a + (-a) = 0 < \sqrt{3a+1}$, which holds for $a > -\frac{1}{3}$. The second inequality becomes $-a < a + \sqrt{3a+1}$, or $0 < 2a + \sqrt{3a+1}$. Let $f(a) = 2a + \sqrt{3a+1}$. Solving $f(a) = 0$ gives $a = -\frac{1}{4}$ (since $a = 1$ is outside the interval). For $a > -\frac{1}{4}$, $f(a) > 0$; for $a < -\frac{1}{4}$, $f(a) < 0$. The condition requires $f(a) > 0$, so $a > -\frac{1}{4}$. Combining with $a \geq -\frac{1}{3}$ gives $-\frac{1}{4} < a < 0$. At $a = -\frac{1}{4}$, $\alpha = \delta$, violating the strict inequality.

Thus the values of a satisfy $-\frac{1}{4} < a < 1$.

Q19. Find all the values of the parameter 'a' for which both roots of the quadratic equation $x^2 - ax + 2 = 0$ belong to the interval $(0, 3)$.

Ans. -

Sol. Hint : $D \geq 0$ & $f(0) \cdot f(3) > 0$ & $0 < (-\frac{b}{2a}) < 3$.

Q20. At what values of 'a' do all the zeroes of the function, $f(x) = (a-2)x^2 + 2ax + a+3$ lie on the interval $(-2, 1)$?

Ans. $a \in (-\infty, -\frac{1}{4}) \cup \{2\} \cup (5, 6]$

Sol. We need all zeros of $f(x) = (a-2)x^2 + 2ax + a+3$ to lie strictly inside the open interval $(-2, 1)$.

Consider the degenerate case when $a = 2$: then the function becomes linear: $f(x) = 4x + 5$, whose sole zero is $x = -\frac{5}{4} = -1.25$, which lies in $(-2, 1)$. Hence $a = 2$ is acceptable.

Now assume $a \neq 2$ so that $f(x)$ is quadratic. Let $A = a-2, B = 2a, C = a+3$. The discriminant is $D = B^2 - 4AC = 4(6-a)$. For real zeros we need $D \geq 0 \Rightarrow a \leq 6$.

The vertex $x_0 = -\frac{B}{2A} = -\frac{a}{a-2}$ must lie inside the interval: $-2 < x_0 < 1$. This gives two inequalities:

$$-\frac{a}{a-2} > -2 \Rightarrow \frac{a-4}{a-2} > 0 \text{ which holds for } a < 2 \text{ or } a > 4;$$

$$-\frac{a}{a-2} < 1 \Rightarrow \frac{1-a}{a-2} < 0 \text{ which holds for } a < 1 \text{ or } a > 2.$$

The intersection of these vertex conditions is $a < 1$ or $a > 4$.

For the zeros to lie strictly inside the interval, the quadratic must not cross the endpoints. Evaluate $f(-2) = a-5$ and $f(1) = 4a+1$. The signs at the endpoints must match the sign of A :

$$A \cdot f(-2) = (a-2)(a-5) > 0 \text{ and } A \cdot f(1) = (a-2)(4a+1) > 0.$$

The first inequality gives $a < 2$ or $a > 5$; the second gives $a < -\frac{1}{4}$ or $a > 2$.

Combine all conditions for the quadratic case: $a \leq 6$, vertex condition ($a < 1$ or $a > 4$), and the two sign inequalities. The intersection is $a < -\frac{1}{4}$ or $5 < a \leq 6$. (Values $a = -\frac{1}{4}$ and $a = 5$ are excluded because they yield zeros exactly at the endpoints.)

Including the linear case $a = 2$, the complete set of values is:

$$a \in \left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6].$$

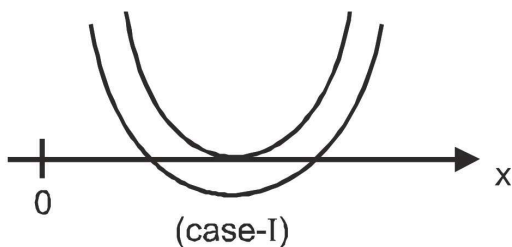
Q21. Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K+5 = 0$ has atleast one positive root.

Ans. $K \in (-\infty, -1]$

Sol. **Case-I** : Both the roots are positive

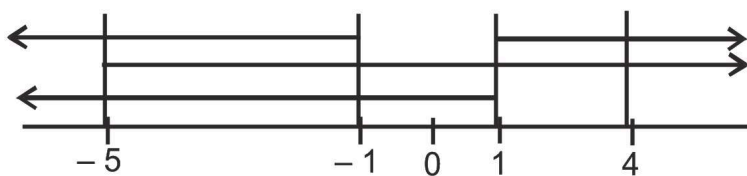
$$x^2 + 2(K-1)x + (K+5) = 0$$

$$(i) \quad D \geq 0 \Rightarrow 4(K-1)^2 - 4(K+5) \geq 0 \Rightarrow (K+1)(K-4) \geq 0$$



$$(ii) \quad f(0) > 0 \Rightarrow K+5 > 0 \Rightarrow K > -5$$

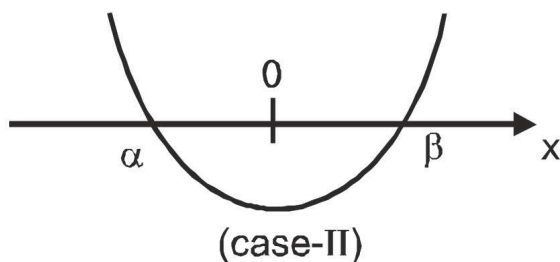
$$(iii) \quad -\frac{b}{2a} > 0 \Rightarrow \frac{2(1-k)}{2} > 0 \Rightarrow K < 1$$



$$K \in (-\infty, -1] \quad \dots (i)$$

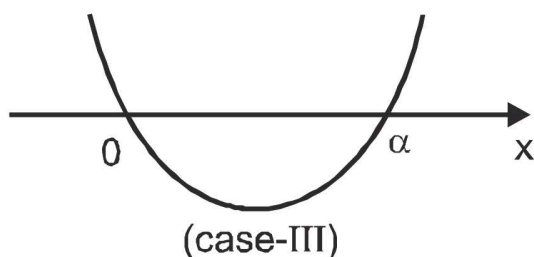
Case-II : One root is +ve and other root is -ve

$$f(0) < 0 \Rightarrow k+5 < 0 \Rightarrow K < -5 \quad \dots (ii)$$



Case-III : One root is zero and other is +ve

$$f(0) = 0 \& \frac{-b}{2a} > 0 \quad \Rightarrow \quad K = -5 \quad \dots \text{(iii)}$$



Union of all the three cases give $K \in (-\infty, -1] \cup (-\infty, -b] \Rightarrow b = 1$. **Ans.**

Q22. If all the solutions of the inequality $x^2 - 6ax + 5a^2 \leq 0$ are also the solutions of inequality $x^2 - 14x + 40 \leq 0$ then find the number of possible integral values of a .

Ans. 0

Sol. We are given two inequalities: $x^2 - 6ax + 5a^2 \leq 0$ and $x^2 - 14x + 40 \leq 0$.

First, solve the second inequality: $x^2 - 14x + 40 \leq 0$.

Factor it: $(x - 4)(x - 10) \leq 0$.

Thus, the solution set is $[4, 10]$.

Now, solve the first inequality: $x^2 - 6ax + 5a^2 \leq 0$.

Its discriminant is $\Delta = 36a^2 - 20a^2 = 16a^2$, so the roots are $\frac{6a \pm 4a}{2}$, i.e., $x = a$ and $x = 5a$.

Thus, the inequality becomes $(x - a)(x - 5a) \leq 0$.

The solution set depends on the sign of a :

- If $a > 0$, then $a < 5a$, so the solution set is $[a, 5a]$.
- If $a < 0$, then $5a < a$, so the solution set is $[5a, a]$.
- If $a = 0$, the inequality is $x^2 \leq 0$, so the solution set is $\{0\}$.

We require that all solutions of the first inequality are also solutions of the second, i.e., the solution set of the first is a subset of $[4, 10]$.

Consider $a = 0$: the solution set is $\{0\}$, which is not contained in $[4, 10]$. So $a = 0$ is invalid.

Consider $a > 0$: we need $[a, 5a] \subseteq [4, 10]$. This gives $a \geq 4$ and $5a \leq 10$, i.e., $a \leq 2$. No a satisfies both, so no positive a .

Consider $a < 0$: the solution set is $[5a, a]$, which consists entirely of negative numbers (since $a < 0$ and $5a < 0$). This cannot be a subset of $[4, 10]$. Hence no negative a works.

Since no real a satisfies the condition, there are no integral values of a .

Thus, the number of possible integral values of a is 0.

Q23. The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

(A) 2 (B) 1 (C) 0 (D) 4

Ans. A

Sol. (i) $x^2 - 8kx + 16(k^2 - k + 1) = 0 \quad \therefore D = 64(k^2 - (k^2 - k + 1)) = 64(k - 1) > 0$

$$\Rightarrow k > 1 \quad \dots\dots(1)$$

$$(ii) -\frac{b}{2a} > 4 \quad \Rightarrow \frac{8k}{2} > 4 \quad \Rightarrow k > 1 \quad \dots\dots(2)$$

$$(iii) f(4) \geq 0$$

$$\Rightarrow 16 - 32k + 16(k^2 - k + 1) \geq 0 \quad \Rightarrow k^2 - 3k + 2 \geq 0$$

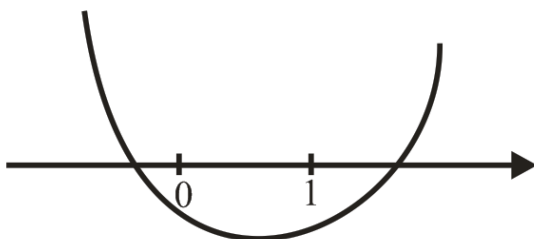
$$\Rightarrow (k - 2)(k - 1) \geq 0 \geq k \leq 1 \text{ or } k \geq 2 \quad \dots\dots(3)$$

(1) $\cap$ (2) $\cap$ (3). Hence $k = 2$

Q24. Largest integral value of 'a' for which the expression $y = x^2 + ax + (a - 3)$ is non positive $\forall x \in [0, 1]$ is

Ans. 1

Sol.



$$(1) D > 0$$

$$a^2 - 4a + 12 > 0$$

$$\Rightarrow (a - 2)^2 + 8 > 0$$

$$\Rightarrow a \in R \quad \dots\dots\dots(1)$$

$$(2) f(0) \leq 0$$

$$a - 3 \leq 0$$

$$a \leq 3 \quad \dots\dots\dots(2)$$

$$(3) f(1)$$

$$\Rightarrow 1 + a + a - 3 \leq 0$$

$$\Rightarrow a \leq 1 \quad \dots\dots\dots(3)$$

(1), (2) and (3)

$$a \leq 1$$

Q25. If the equation $x^4 - ax^3 + 11x^2 - ax + 1 = 0$ has four distinct positive roots then the range of a is (m, M). Find the value of $m + 2M$.

Ans. 19

Sol. $x^4 - ax^3 + 11x^2 - ax + 1 = 0$

divide by x^2

$$x^2 - ax + 11 - \frac{a}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) - a\left(x + \frac{1}{x}\right) + 11 = 0$$

put $x + \frac{1}{x} = t$

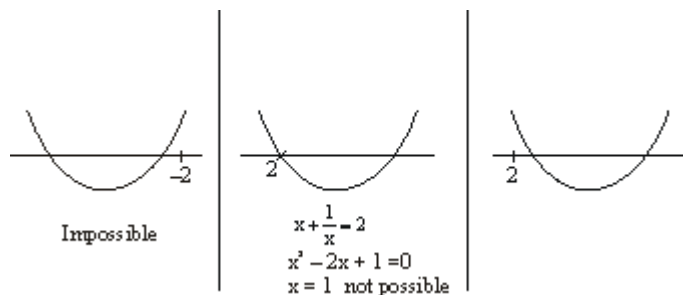
$$x^2 + \frac{1}{x^2} = t^2 - 2$$

$$t^2 - 2 - at + 11 = 0$$

$$t^2 - at + 9 = 0$$

$$x + \frac{1}{x} \leq -2$$

$$x + \frac{1}{x} \geq 2$$



$$x + \frac{1}{x} = 2$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0 \text{ \{two equal roots\}}$$

$$x = 1 \text{ Not possible}$$

$$t^2 - at + 9 = 0$$

$$t_1, t_2 \text{ are the roots.}$$

$$(i) D > 0$$

$$a^2 - 36 > 0$$

$$(a + 6)(a - 6) > 0$$

$$a \in (-\infty, -6) \cup (6, \infty)$$

$$(ii) \frac{-b}{2a} > 2$$

$$\frac{a}{2} > 2$$

$$a > 4$$

$$a \in (4, \infty)$$

$$(iii) a \cdot f(2) > 0$$

$$1 \times (4 - 2a + 9) > 0$$

$$2a - 13 < 0$$

$$a < \frac{13}{2}$$

$$a \in \left(-\infty, \frac{13}{2}\right)$$

(i), (ii), (iii)

$$a \in \left(6, \frac{13}{2}\right)$$

$$M = 6, M = \frac{13}{2}$$

$$\Rightarrow m + 2M = 6 + 2 \cdot \frac{13}{2} = 19$$

Q26. When $y^2 + my + 2$ is divided by $(y - 1)$ then the quotient is $f(y)$ and the remainder is R_1 . When $y^2 + my + 2$ is divided by $(y + 1)$ then quotient is $g(y)$ and the remainder is R_2 . If $R_1 = R_2$ then find the value of m .

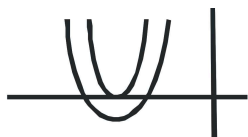
Ans. 0

Sol. -

Q27. Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.

Ans. -

Sol.



$$x^2 + 2(K - 1)x + K + 5 = 0$$

$$D > 0$$

$$\Rightarrow 4(K^2 - 3K - 4) \geq 0 \Rightarrow (K - 4)(K + 1) \geq 0$$

$$\Rightarrow K \in (-\infty, -1) \cup (4, \infty)$$

Now subtract those cases in

which both roots are negative

$$(i) \quad f(0) > 0 \Rightarrow k + 5 > 0 \Rightarrow K > -5$$

$$(ii) \quad -\frac{b}{2a} < 0 \Rightarrow 1 - k < 0 \Rightarrow k > 1$$

So for $K > 1$ both roots are negative

hence for at least one positive root $K \in (-\infty, -1)$

Q28. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$

Ans. 20

Sol. $x^2 + 18 + 45 - 2\sqrt{x^2 + 18 + 45} + 1 = 16$

$$\Rightarrow \left(\sqrt{x^2 + 18x + 45} - 1 \right)^2 = 16$$

$$\Rightarrow \sqrt{x^2 + 18x + 45} - 1 = 4$$

$$\Rightarrow \sqrt{x^2 + 18x + 45} = 4 + 1 = 5, -3$$

$$\Rightarrow x^2 + 18x + 45 = 25, \text{ (Reject } -3)$$

$$\Rightarrow x^2 + 18x + 20 = 0$$

Product of root = +20.

Q29. Find all real numbers x such that, $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$.

Ans. -

Sol. -

Q30. Solve : $(x + 9)(x - 3)(x - 7)(x + 5) = 385$

Ans. $-4, 2$ and $-1 \pm \sqrt{71}$

Sol. $(x + 9)(x - 3)(x - 7)(x + 5) = 385$

$$(x + 9)(x - 7)(x - 3)(x + 5) = 385$$

$$(x^2 + 2x - 63)(x^2 + 2x - 15) = 385$$

$$\text{put } x^2 + 2x = t$$

$$(t - 63)(t - 15) = 385$$

$$t^2 - 15t - 63t + 945 - 385 = 0$$

$$t^2 - 78t + 560 = 0$$

$$t^2 - 70t - 8t + 560 = 0$$

$$(t - 70)(t - 8) = 0$$

$$t = 70 \text{ and } 8$$

$$\text{for } t = 70$$

$$x^2 + 2x - 70 = 0$$

$$x = \frac{-2 \pm \sqrt{4 + 4 \times 70}}{2}$$

$$x = -1 \pm \sqrt{71}$$

$$\text{for } t = 8$$

$$x^2 + 2x - 8 = 0$$

$$x^2 + 4x - 2x - 8 = 0$$

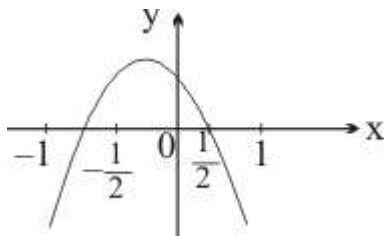
$$x(x + 4) - 2(x + 4) = 0$$

$$(x + 4)(x - 2) = 0$$

$$x = -4, 2$$

Paragraph

Let $f(x) = ax^2 + bx + c$ be a quadratic expression and $y = f(x)$ has graph as shown in figure



Q31. Which of the following is false ?

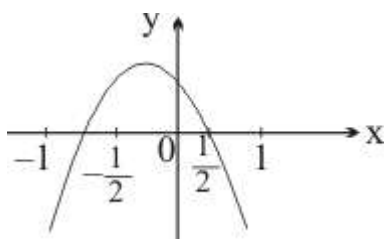
- (A) $ab > 0$ (B) $abc < 0$ (C) $ac < 0$ (D) $bc < 0$

Ans. 5ed8ce92c45aa173eef23b1b

Sol. -

Paragraph

Let $f(x) = ax^2 + bx + c$ be a quadratic expression and $y = f(x)$ has graph as shown in figure



Q32. Which of the following is true ?

- (A) $a + b + c > 0$ (B) $a - b + c > 0$ (C) $a + 3b + 9c < 0$ (D) $a - 3b + 9c > 0$

Ans. 5ed8cf02b8e12773f3b6395c

Sol. -

Paragraph

Consider a rational function $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ and a quadratic function $g(x) = x^2 - (m + 1)x + m - 1$, where m is a parameter.

Q33. Number of integral value(s) of ' m ' so that $g(x)$ is always positive, is

- (A) 0 (B) 1 (C) 2 (D) more than 2

Ans. 5ed8d05eb8e12773f3b6398b

Sol. -

Paragraph

Consider a rational function $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ and a quadratic function $g(x) = x^2 - (m + 1)x + m - 1$, where m is a parameter.

Q34. Number of integral values in the range of $f(x)$, is

- (A) 1 (B) 2 (C) 3 (D) more than 3

Ans. 5ed8d110c45aa173eef23b8a

Sol. -

Paragraph

Consider a rational function $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ and a quadratic function $g(x) = x^2 - (m + 1)x + m - 1$, where m is a parameter.

Q35. If both roots of $g(x) = 0$ are greater than the smallest value of the function $f(x)$, then 'm' lies in the interval

- (A) $(-\infty, -2)$ (B) $\left(-\infty, \frac{1}{4}\right)$ (C) $(-2, \infty)$ (D) $\left(-\frac{1}{2}, \infty\right)$

Ans. 5ed8d3344f8dba740ac55b1e

Sol. -

Paragraph

For $a, b \in \mathbb{R} - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in \mathbb{R}$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

\n

Q36. The value of $(a + b)$ is equal to

- (A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. -

Paragraph

For $a, b \in \mathbb{R} - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in \mathbb{R}$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

\n

Q37. The minimum value of $f(x)$ in $\left[0, \frac{3}{2}\right]$ is equal to

- (A) $-\frac{33}{8}$ (B) 0 (C) 4 (D) -2

Ans. 5ed8d7a74f8dba740ac55b25

Sol. -

Q38. The value(s) of 'p' for which the equation $ax^2 - px + ab = 0$ and $x^2 - ax - bx + ab = 0$ may have a common root, given a, b are non zero real numbers, is :

- (A) $a + b^2$ (B) $a^2 + b$ (C) $a(1 + b)$ (D) $b(1 + a)$

Ans. B, C

Sol. Equation $ax^2 - px + ab = 0$... (i)

$$x^2 - ax - bx + ab = 0$$

from equation, $x = a, b$

$\therefore$ both equations may have a common root.

for $x = a$

$$a^3 - ap + ab = 0$$

$$a(a^2 - p + b) = 0$$

$$a \neq 0, p = a^2 + b$$

for $x = b$

$$ab^2 - pb + ab = 0$$

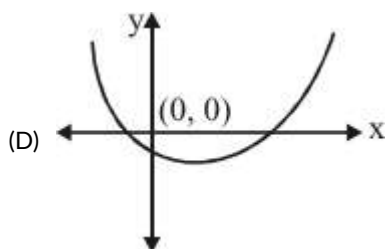
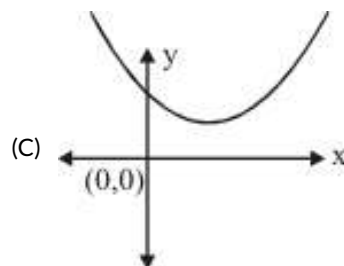
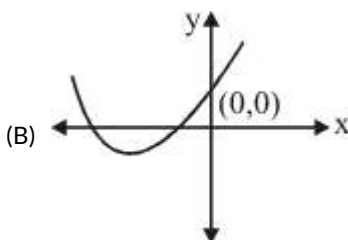
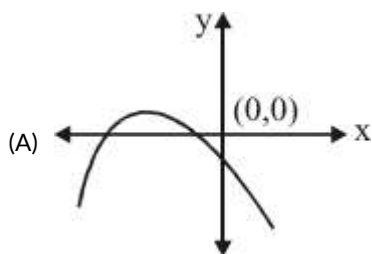
$$b(ab - p + a) = 0$$

$$b \neq 0$$

$$p = a(1 + b)$$

Option B, C are correct.

Q39. For which of the following graphs of quadratic expression $y = ax^2 + bx + c$, then product abc is positive?

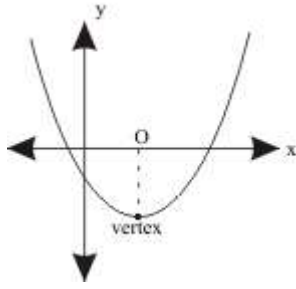


Ans. B, D

Sol. Option (B) $a > 0, b > 0$ and $c > 0$, then $abc > 0$.

Option (D) $a > 0, b < 0$ and $c < 0$, then $abc > 0$.

Q40. Graph of $y = ax^2 + bx + c = 0$ is given adjacently. What conclusions can be drawn from this graph-



- (A) $a > 0$ (B) $b < 0$ (C) $c < 0$ (D) $b^2 - 4ac > 0$

Ans. D, B, C, A

Sol. -

Q41. The values of a for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ does not have a real solution is

- (A) -10 (B) 12 (C) 5 (D) -30

Ans. D, C, B

Sol. -

Q42. If α, β are the roots of $4x^2 - 16x + c = 0$, $c > 0$ such that $1 < \alpha < 2 < \beta < 3$, then value of ' c ' can be

- (A) 11 (B) 12 (C) 13 (D) 14

Ans. D, C

Sol. -

Q43. If the quadratic polynomial $P(x) = (p - 3)x^2 - 2px + 3p - 6$ ranges from $[0, \infty)$ for every $x \in \mathbb{R}$, then the value of p can be

- (A) $\frac{3}{2}$ (B) 4 (C) 6 (D) 7

Ans. C

Sol. -

Q44. If $f(x) = 2x^2 + kx + 50$, $k \geq 20$ and magnitude of difference of the roots of the equation $f(x) = 0$ is 8, then which of the following can be true -

- (A) minimum value of $f(x)$ is -32 (B) minimum value of $f(x)$ is -64
 (C) graph of $f(x)$ cuts positive y-axis (D) graph of $f(x)$ cuts negative y-axis

Ans. A, C

Sol. -

- Q45. Let α and β ($\alpha > \beta$) be roots of the quadratic equation $ax^2 + bx + c = 0$.
If $0 < a < c$ and $a + c = b$, then which of the following statement(s) is/are correct?
(A) Larger root is 1. (B) Larger root is -1 . (C) Smaller root is $\frac{c}{a}$.
(D) b is greater than zero.

Ans. B, D

Sol. -

- Q46. Consider the quadratic trinomial $f(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter. Find the range of p in the following conditions given in column-I.

Column-I

Column-II

- (A) If both roots of $f(x) = 0$ are confined in $(-1, 1)$ then (P) $\left(\frac{2}{5}, \infty\right)$
(B) Exactly one root of $f(x) = 0$ lies in $(-1, 1)$ (Q) ϕ
(C) Both roots of $f(x) = 0$ are greater than 1 (R) $\left(-\frac{1}{17}, \frac{1}{3}\right)$
(D) One root of $f(x) = 0$ is greater than 1 and other root of (S) $\left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$
 $f(x) = 0$ is less than -1 (T) $\left(-\infty, -\frac{1}{17}\right) \cup \left(\frac{1}{3}, \infty\right)$

- (A) A - R, B - S, C - Q, D - Q (B) A - R, B - S, C - T, D - Q

Ans. 5ed8dd0126f1a4740560799b

Sol. -

- Q47. The roots of the quadratic equation, $x^2 - 5x + 6 = 0$ is α and β . Then find the following.

Column I

Column II

- (A) $\alpha - \beta$ (P) 1
(B) $\alpha^2 + \beta^2$ (Q) -13
(C) $\frac{\alpha^3 + \beta^3}{35}$ (R) -1
(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3}$ (S) 13

- (A) (A) -PR (B) - PS (C) - P (D) - S (B) (A) -PR (B) - S (C) - P (D) - P
(C) (A) -PS (B) - SP (C) - P (D) - P (D) (A) -PR (B) - RS (C) - S (D) - P

Ans. 5ed8e526b8e12773f3b6399c

Sol. -

Q48. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Column I

- (A) If $-1 < x < 1$, then $f(x)$ satisfies
- (B) If $1 < x < 2$, the $f(x)$ satisfies
- (C) If $3 < x < 5$, then $f(x)$ satisfies
- (D) If $x > 5$, then $f(x)$ satisfies

Column II

- (P) $0 < f(x) < 1$
- (Q) $f(x) < 0$
- (R) $f(x) > 0$
- (S) $f(x) < 1$

(A) (A) - PRS (B) - Q (C) - QS (D) - PR (B) (A) - PR (B) - QS (C) - QS (D) - PR

(C) (A) - PRS (B) - QS (C) - QS (D) - PRS (D) (A) - QS (B) - PS (C) - QS (D) - PRS

Ans. 5ed9c480b8e12773f3b63ed9

Sol. -